

Evaluating the impact of an application dedicated to recycling highly polluting materials while integrating environmental education in Puebla.

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Abstract. The purpose of the project is to evaluate the impact of using a mobile application designed to promote the recycling of highly polluting waste in the city of Puebla, as well as its relationship with citizens' environmental awareness. The initiative stems from the need to provide an accessible tool that informs users about the recycling centers and, at the same time, promotes sustainable practices through environmental education.

The research will be conducted using a mixed-method approach, with a quantitative focus, employing techniques such as surveys and interviews to gather information on participants' current recycling habits and environmental awareness. Subsequently, a functional mobile application will be designed and tested to facilitate the location of recycling points. The results will allow for an analysis of the effectiveness of the technological tool in promoting greater citizen participation in recycling and reducing the environmental impact caused by hazardous waste.

Key words: Highly polluting waste, application, recycling rate, Puebla, recycling centers

1 Introduction

The transition towards circular economy models has become a cornerstone of global sustainable development. Within this paradigm, recycling evolves from a mere waste management alternative to a key strategy to conserve and protect natural resources, as well as the environment. However, this process faces challenges when dealing with highly polluting waste, whose toxic characteristics and complex handling processes demand specialized solutions and an informed, active citizenry.

Information technologies, particularly mobile applications, have emerged as powerful tools to facilitate interaction between citizens and urban services, democratizing access to information and promoting positive behavioral changes. Their application in the environmental field has shown potential for bridging the gap between the intention to recycle and concrete action, a recurring problem in many cities.

This project is situated at the intersection of these two realities: the urgent need to improve hazardous waste management and the potential of mobile technology as a catalyst for change. This protocol defines the methodology for investigating the concrete impact of a technological intervention in the specific context of the city of Puebla, Mexico. The study aims to contribute to a knowledge gap that is both geographical —being the first of its kind in the region— and conceptual, by integrating logistical management with environmental education within a single platform.

2 Problem statement and setting

In the city of Puebla, the increase in the generation of highly polluting waste (appliances, batteries, and oil) represents a challenge: on the one hand, improper disposal contaminates soil, water bodies, and drainage networks; on the other, these materials and opportunities to give them a second chance are wasted. Although there are collection centers and specific campaigns in the city, the rate of formal collection and specialized treatment remains insufficient given the actual volume of waste generated by households and businesses.[1]

In Puebla, the generation of municipal solid waste is high: the capital municipality produces approximately 179,730 tons/month. In the absence of statistics by type of waste, and based on local campaigns and conservative assumptions about waste composition, it is estimated that electronic waste could represent between 7,200 and 21,560 tons/year (1–3% of municipal solid waste), used oil between 718 and 2,156 tons/year (0.1–0.3% of municipal solid waste), and batteries, although light in weight, could amount to hundreds of tons per year if urban sources are considered (estimate: 359 tons/year assuming 0.05% of municipal solid waste). These figures require validation with local measurements (mapping of collection points and scales in campaigns), but they allow us to gauge the scale of the problem and justify the solution to it.[4]

In Puebla, campaigns and collection points for different types of waste have been promoted: the City Council publishes information on municipal collection centers where electronics and other materials are accepted; the Benemérita Universidad Autónoma de Puebla (BUAP) runs campaigns and programs for the collection of used vegetable oil and promotes electronic recycling in the university community, and specialized private organizations/companies also operate collection and recycling channels for household appliances in the region[5]. However, efforts are fragmented (temporary or institutional campaigns) and there is not always a reliable and accessible route for all citizens to dispose of their batteries, oils, and household appliances safely.

The following shows the impact that this waste has on both health and the environment:

- Batteries: contain heavy metals (e.g., mercury, cadmium, lead in older alkaline types) which, if they reach uncontrolled landfills or soil/water, pose a risk of bioaccumulation and toxic effects. The lack of a stable municipal collection program increases the likelihood of improper disposal. [2]
- Vegetable oil and lubricants: domestic cooking oil disposal damages sewer systems and pollutes water bodies; local collection alternatives (e.g., Agua de Puebla campaigns and university collections) show that collection is viable, and waste can be converted (biodiesel, soap, etc.), but citizen coverage is still limited. [3]
- Appliances: Informal drying and inadequate separation hinder the recovery of metals and components and encourage dangerous practices (informal dismantling or burning). The presence of collection centers and companies that receive appliances is a starting point, although greater coordination between the government, private initiative, and citizens is necessary.[1]

2.1 Lack of knowledge

Analysis of existing literature on mobile applications for promoting recycling reveals three critical gaps. First, there is a limitation in the scope of materials, as technological solutions and associated studies tend to specialize in a single type of waste. This leaves unaddressed the need for a tool that manages multiple highly polluting waste streams in an integrated manner.

This shortcoming is exacerbated in the local geographical context, as no previous research or similar technological developments have been identified that address the issue of recycling through a mobile application in the city of Puebla, representing a clear knowledge gap at the regional level.

Additionally, a gap in functional and educational focus has been identified. While various applications promote recycling, few studies analyze platforms explicitly designed as educational and awareness-raising tools. The literature lacks evidence on systems that integrate educational content to cultivate a deep understanding among users of best practices and the environmental impact of their decisions.

2.2 Context

The study will be conducted in the city of Puebla, where recycling of highly polluting materials is limited due to insufficient information, poor environmental education, and a lack of technological tools to facilitate this practice. The research will focus on users interested in protecting the environment and promoting sustainable habits, specifically covering materials such as batteries, used oil, and household appliances, which represent a significant source of pollution when not managed properly. This context of misinformation and lack of tools in Puebla takes on a critical dimension when analyzing the quantitative magnitude of the problem.

2.3 Magnitude

The population of the city of Puebla in 2025 is estimated to be approximately 3,443,290 inhabitants, with a growth rate of 1.44%. Statistics show that in 2024, approximately 9,000 tons of waste were generated per day in the city of Puebla, however, only about 8% was recycled[9]. In 2025, the amount of waste generated per day tripled[8], but this does not mean that the culture and level of recycling has increased. On the contrary, it has remained the same, which indicates a major problem for Puebla, referring to all types of waste, most of which are common. Now, in terms of highly polluting materials, 9% was recycled in 2023. Currently, there is no exact figure for the amount recycled, the reason being that these are the least recycled materials. Considering that the culture of recycling has not advanced, the recycling of highly polluting materials has not advanced either. The large volume of waste generated daily translates, in turn, into a constant and cumulative frequency of improper disposal.

2.4 Frequency

The problem of inadequate management of highly polluting waste occurs on a daily basis and on a considerable scale, as shown by the following statistics:

- Electronic waste: Nationwide, 1.2 million tons are generated annually, with a per capita average of 9.9 kg (UNDP, 2021). For the metropolitan area of Puebla, with more than 3 million inhabitants, this would represent a daily and constant generation of approximately 81 tons of this hazardous waste. [6]
- Batteries: Each year, an estimated 990 million units are discarded in Mexico. This means that, on average, more than 1,800 batteries are thrown away every minute without proper handling, ensuring continuous contamination of soil and aquifers. [7]

- Cumulative Nature: The consistency of these figures shows that the problem is not sporadic. The daily frequency implies a progressive and uninterrupted accumulation of pollutants in the local ecosystem, underscoring the urgency of a permanent and accessible solution for citizens.[7]

The persistent and large-scale nature of this problem makes it necessary to investigate its underlying causes.

2.5 Problem causes

Based on the problem we want to solve, it is clear that not recycling these types of materials has a major environmental impact, which represents a critical problem in society. In relation to this, the main causes of this problem are listed below, explaining why these materials are not recycled:

- Lack of information on how to recycle materials properly.
- Lack of knowledge about the location of collection centers.
- Lack of environmental awareness in society.
- Limited market for recycled materials.

2.6 Possible solutions

In response to the identified causes, such as lack of information, disconnect with recycling centers, and low recycling rates for these three materials, the proposed solution is to design and develop a mobile application available for any operating system, with geolocation tools that act as a link between the user and the recycling center specializing in a particular material. The application would show the user's location and the location of the recycling center on an interactive map, along with information such as opening hours and days, whether there is any remuneration from the recycling center, and how the materials should be prepared before taking them there. It will also have an educational section.

2.7 Unanswered questions

- Is the lack of recycling technologies the main barrier to managing highly polluting materials, or are factors related to limited public awareness the real obstacle in Puebla?
- How can we assure users that the materials collected through the app will be properly treated and recycled, rather than simply being transferred to another landfill or handled through informal and dangerous channels?

3 Justification

The low level of recycling of highly polluting waste (mainly batteries, electronic devices, used oil, and other special waste) in the city of Puebla represents a significant environmental and public health problem, the scale of which transcends the municipal sphere. When disposed of improperly, these materials generate toxic emissions that directly impact soil quality, water (including crucial watersheds such as the Atoyac River and its regional impact), and air. Their improper management not only accelerates environmental deterioration in the region but also contributes to the accumulation of toxic pollutants that affect the public health of the population of Puebla.

This situation is chronically aggravated by a double gap: a lack of environmental awareness and the absence of accessible, timely, and geolocated information on collection points and specialized recycling centers. This perpetuates inappropriate waste disposal habits. Therefore, we seek to generate original and relevant knowledge by improving access to information. This guides us in the design and implementation of a mobile application as a tool for environmental awareness and connection between recycling centers and the population of Puebla.

By focusing on highly polluting waste, environmental and health benefits are maximized, reducing the exposure of flora, fauna, and the population to harmful elements such as heavy metals. This leads to improved public health through the reduction of pollutants, a decrease in toxic load, the strengthening of recycling, and the promotion of citizen participation.

The results of this project will be disseminated to the Puebla community in a practical way through the application itself, as part of the environmental awareness that it seeks to generate and convey.

4 Objectives

The general objective of this project is evaluating the impact of an application dedicated to recycling highly polluting materials with respect to recycling levels, as well as its contribution to environmental education and awareness within the context of the city of Puebla.

4.1 Specific objectives

- Identify the recycling rates of highly polluting materials processed at collection centers in Puebla and analyze how citizens' environmental awareness influences these rates.
- Implement a functional mobile application that allows users to view recycling centers near their location, while also providing educational information to foster environmental awareness and good recycling practices at home.
- Compare the results of using the application for a specific period of time in the city of Puebla by collecting and analyzing data on the frequency of use of the application and the volume of highly polluting waste recycled, taking into account the data collected before its implementation.

4.2 Research questions

What are the current recycling rates for highly polluting materials at collection centers in Puebla, and how do they relate to citizens' level of environmental awareness?

What are the key features and functionalities that a mobile app should have to make it easier for users to locate recycling centers and effectively promote recycling practices at home in Puebla?

Has there been a significant change in the volume of highly polluting waste recycled and in the frequency of recycling in Puebla after the implementation and use of the mobile app, compared to previous data?

5 Methodology approach

The research will be conducted using a mixed approach, with qualitative and quantitative data collection techniques applied to highly polluting waste, with the quantitative methodology being the dominant one. Questionnaires will be used to measure the current impact on recycling centers, and surveys will also be conducted to gather descriptive and detailed information.

5.1 Sampling

Stratified probability sampling will be used for data collection. Given that the population of Puebla is heterogeneous in terms of characteristics that could significantly influence the use of the application and recycling practices, this design will ensure that the various key subgroups are represented within the total sample.

The population will be divided into strata based on two main criteria:

- Geographic Area: To capture differences in access to services and infrastructure, a distinction will be made between urban and rural populations.
- Age range: To measure environmental awareness across generations, groups will be established based on age ranges.

Once the strata have been defined, a random sample will be selected within each of them, proportional to their size in the general population. This strategy will not only allow for accurate estimates for Puebla as a whole, but will also facilitate subsequent analytical comparison of the impact of the app among the different strata.

6 Intervention proposal

To implement and evaluate this proposal, the steps to be taken to meet the proposed objectives are detailed below:

Preparation and design

- I. Gather information on current recycling levels and environmental awareness.
- II. Select the sample where the project will initially be tested
- III. Develop the initial surveys and questionnaires
- IV. Define the key features of the mobile application (all the options it should include)

Implementation and execution

- I. The mobile application will be developed based on all the previous characteristics that were defined in the initial stage.
- II. The necessary tests will be created for the application before it is officially released.
- III. Once everything has been checked and verified, the application will be officially released in all online stores for download (Google Play Store, Apple App Store).
- IV. The application metrics will be monitored (number of downloads, amount of recycled material).

Evaluation and analysis

- I. A questionnaire will be administered again to the same initial sample to measure changes in knowledge, attitudes, and habits.
- II. The final data from the application will be collected.
- III. The results of the final survey will be compared with the initial survey to determine whether there was a significant change.
- IV. The app data will be analyzed to assess the level and impact it had on the amounts of material recycled.
- V. Based on the findings, a conclusion will be reached as to whether or not the app was effective in meeting the objectives set out, and recommendations for future improvements will be offered.

7 Ethical issues

The development of this project will strictly adhere to ethical research principles.

1. Informed Consent: Participation in all stages of data collection (such as surveys and questionnaires) will be voluntary. Before applying any instrument, participants will be provided with a clear explanation of the study's objective, the use to which the information will be put, and the approximate time required. Their explicit consent will be requested, assuring them that they may withdraw from the study at any time without penalty.

2. Confidentiality and Anonymity: The privacy of all participants will be protected. Data collected through surveys will be handled anonymously and confidentially. No personally identifiable information (such as names or exact addresses) will be requested, and the information will be stored in secure, encrypted databases accessible only by the research team.
3. Application Data Management: In relation to the mobile application, users will be informed that the geolocation functionality is used exclusively to identify nearby collection centers in real time and will not involve the storage, tracking, or monitoring of the user's personal location.
4. Integrity and Truthfulness: The project is committed to honesty in the handling and reporting of data. The results will be analyzed and presented objectively, without falsifying or altering the information obtained, whether the findings confirm or refute the initial hypotheses.
5. Non-maleficence: Care will be taken to ensure that the design of the instruments (surveys) does not contain invasive, sensitive, or offensive questions that could cause discomfort or psychological harm to participants. The benefits of the project, focused on environmental improvement and education, will always seek to outweigh any minimal risks associated with participation.

8 State of the art

Fifteen documents were analyzed, including theses, research articles, and degree projects.

The research comes from various Latin American countries, mainly:

- Colombia (Bogotá, Ocaña)
- Peru (Huancavelica, Cajamarca, Lima)
- Ecuador (Guayaquil, Quito)
- Mexico (Sonora)
- Argentina (Córdoba)

The sources were selected and analyzed for their relevance to the development of mobile applications focused on recycling. The areas of analysis were:

Functionality: Whether the app focuses on education, logistics (connection), gamification (incentives), or a combination.

- Type of waste: Whether they focus on general municipal solid waste (MSW) (PET, cardboard, glass) or specific waste (electronic, organic).
- Methodology: The research approach (quantitative, qualitative, mixed) used to validate the proposal.
- Publication date: Articles published in or after 2020 were included, as technology and application development evolve rapidly over time. For the state of the art to be relevant, it is crucial to analyze solutions and platforms developed in the technological context of the last 5 years.

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A marked tendency was detected to focus apps on high-turnover, low-risk waste (PET, cardboard, as in sources 2, 3, 4, 11) or on a single stream of special waste (electronics, as in sources 7 and 8). None of the sources analyzed propose an integrated solution for multiple types of highly polluting waste (batteries, oil, and electronics), which represents the main gap that this project seeks to fill.

Analysis of the results from the 15 sources reveals several consistent patterns and findings regarding the implementation of recycling apps:

- Incentives as the Main Driver: A majority of studies (Sources 3, 4, 6, 9, 11, 13) conclude that incentives (digital coins, points, discounts, money) are a critical factor in motivating participation and adoption of the app.

- **The Gap Between Interest and Knowledge:** A recurring disconnect is identified: citizens express interest in recycling (Source 4: 88.34% interest), but report a high level of ignorance about how to do it correctly (Source 12) or who collects the material (Source 1: 51% do not know the collector).
- **Education as a Necessity:** Several sources (2, 5, 10, 14, 15) propose educational modules, educational games, and tutorials, recognizing that the app should serve as a training tool to close the knowledge gap.
- **Mixed Results on Effectiveness:** Evidence on direct impact is mixed. While one study (Source 15) claims that an app improved waste segregation, another (Source 7, on electronic waste) reported no significant difference in knowledge after 5 months of use, suggesting that usability or lack of underlying education may be barriers.
- **The Importance of Logistics:** Mistrust and logistical problems are a key barrier. Users express frustration when the materials they have separated are mixed again by collectors (Source 1) and demand regular service and identifiable collectors.

The literature confirms the educational gap we seek to address (Sources 1, 12). However, unlike studies that prioritize external incentives (money, points), our project will focus on environmental awareness and logistical ease. Instead of creating a new system, we will make the existing one visible through geolocation, addressing logistical ignorance (Source 1) and measuring the impact of education, not gamification.

The context in which the studies analyzed were conducted is essential to understanding their scope. It was found that all studies were conducted in urban contexts in Latin American cities (Bogotá, Lima, Quito, Guayaquil, Cajamarca). This validates the feasibility of implementing technological solutions to MSW problems in environments with high population density and smartphone penetration.

Although the context is similar, the profile of the participants varies:

- **General Public/Citizens:** Most surveys are aimed at the general public (Sources 1, 3, 4, 9, 13, 15).
- **Age Segmentation:** The younger population (18-34 years old) is the most receptive and active user of technology (Sources 1, 12), although the projects seek to include all ages (Sources 5, 15).
- **Specific actors:** Some studies focus on key actors in the chain, such as professional collectors (Source 1) or students (Sources 5, 7, 12).

The review confirms a key geographical and contextual gap: no similar studies, theses, or applications addressing the issue of recycling through an app in the city of Puebla, Mexico, were identified.

Studies in other capitals (Sources 1, 10, 13) prove that the technological solution is regionally viable.

Our project will bridge the geographical gap by being the first to be implemented in Puebla. In addition, we will use stratified probability sampling, a more robust methodology than that used by other sources (Source 7), to obtain data that is more representative of our city.

Analysis of the methodological approaches used in the sources is crucial to defining the research design.

Quantitative research is the predominant approach (Sources 4, 6, 7, 8, 12, 13, 15). It focuses on measuring willingness to use, perception, and impact through numerical values.

However, several of the most robust studies (Sources 1, 3, 14) use a mixed approach. They combine quantitative data collection (surveys) with qualitative techniques (interviews, observation) to understand the “why” behind the numbers, especially when dealing with different actors (citizens vs. collectors).

Several of the articles used techniques such as surveys and questionnaires (Sources 1, 3, 4, 7, 8, 12, 13, 14, 15), often distributed via Google Forms, to measure perception, knowledge, and intention. Studies that evaluate impact (and not just intention) use pre-experimental pre-test and post-test designs (Sources 6, 7, 8, 15) to measure change in knowledge or recycling volumes before and after implementing the app. Attitudes and perceptions (Sources 7, 8, 13) were measured using the Likert scale, mainly in questionnaires.

Literature validates our methodological design. The mixed approach (Sources 1, 3, 14) and the use of surveys (Sources 3, 4, 8, 12) are consistent with the standard for these studies. Fundamentally, our intervention proposal replicates the successful pre-test/post-test design (Sources 6, 8, 15) to objectively measure the impact of the application.

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Annex

Questions for center

1. What types of highly polluting materials (batteries, oils, electronics) does your center accept?
2. What is the average volume (in kg) you receive for each material per month?
3. On a scale of 1 to 5, how has the volume received changed over the last 12 months?
4. Who are your main suppliers of these materials? (Individual citizens, businesses, educational institutions, others).
5. Approximately how many people drop off materials at your center per week?
6. What are your public operating hours and days?
7. Do you offer any type of incentive (economic or otherwise) for dropping off materials?
8. Do you have the capacity to increase the volume of material received by [X]%?
9. On a scale of 1 (Strongly Disagree) to 5 (Strongly Agree), please rate the following statements:
 - a) "The main barrier to receiving more material is the public's lack of knowledge about our existence."
 - b) "A mobile application that geolocates us would significantly increase traffic to our center."
 - c) "We would be willing to collaborate by providing official information about our services for such an application."

User

1. What do you currently do with used batteries, cooking oil, or old appliances? Why?
2. When you think of "recycling," what types of materials come to mind first?
3. If you don't recycle these materials, what do you believe are the main reasons? (Explore: lack of information, distance to centers, distrust, inconvenience, etc.).
4. If you have tried to recycle them, what was your experience? Did you encounter any difficulties?
5. How do you usually search for information about recycling topics or local services?
6. Imagine an app that shows you the nearest centers on a map for dropping off each material, including their hours and instructions. Would you use it?
7. What information would be most useful for this application to have? (e.g., how to prepare materials, if incentives are offered, if the center is trustworthy).
8. Beyond location, what other functionality would motivate you to use the app? (e.g., reminders, educational information about the impact, a reward system).

9. What would motivate you more to change your habits: knowing the concrete environmental benefit, receiving an economic reward, or simply the ease of knowing how and where to do it?
10. What would make you trust a collection center listed in an app more?

Instrument for test

Center

1. Compared to the period before the app was launched, have you noticed a change in the volume of highly polluting materials received?
 - a) Significant decrease
 - b) No change
 - c) Slight increase
 - d) Significant increase
2. Approximately what percentage of people who drop off materials mention that they learned about your center through the app?
 - a) 0-10%
 - b) 11-30%
 - c) 31-50%
 - d) More than 50%
3. On a scale of 1 to 5, how positive has the impact of the application been on your workflow and visibility?
(1 = Very unfavorable, 5 = Very favorable)
4. Would you recommend other recycling centers to join this platform?"
 - a) Definitely yes
 - b) Probably yes
 - c) Neutral
 - d) Probably no
 - e) Definitely no

Test User

1. Describe the part of the app that you found most useful and why.
2. What obstacles did you encounter when using the app to locate and get to a collection center?